

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                    | Marks     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 3(a)     | an arrow horizontally on the page to the left labelled $F$                                                                                                                                                                                                                                                                                                                                | <b>B1</b> |
|          | two arrows horizontally on the page to the right each labelled $T$                                                                                                                                                                                                                                                                                                                        | <b>B1</b> |
| 3(b)(i)  | $T = 110 / 2$<br>$= 55 \text{ N}$                                                                                                                                                                                                                                                                                                                                                         | <b>A1</b> |
| 3(b)(ii) | $x = F / k$                                                                                                                                                                                                                                                                                                                                                                               | <b>C1</b> |
|          | $= 110 / 250$<br>$= 0.44 \text{ m}$                                                                                                                                                                                                                                                                                                                                                       | <b>A1</b> |
| 3(c)(i)  | extension = $55 / 250$ <b>or</b> $110 / (2 \times 250)$                                                                                                                                                                                                                                                                                                                                   | <b>C1</b> |
|          | extension = $0.22 \text{ m}$                                                                                                                                                                                                                                                                                                                                                              | <b>A1</b> |
| 3(c)(ii) | $E = \frac{1}{2} Fx$ <b>or</b> $E = \frac{1}{2} kx^2$ <b>or</b> $E = \frac{1}{2} F^2 / k$                                                                                                                                                                                                                                                                                                 | <b>C1</b> |
|          | $E_1 / E_2 = (\frac{1}{2} \times 110 \times 0.44) / (2 \times (\frac{1}{2} \times 55 \times 0.22))$<br><b>or</b><br>$E_1 / E_2 = (\frac{1}{2} k \times 0.44^2) / (2 \times (\frac{1}{2} k \times 0.22^2))$<br><b>or</b><br>$E_1 / E_2 = (\frac{1}{2} \times 110^2 / k) / (2 \times (\frac{1}{2} \times 55^2 / k))$<br>$E_1 / E_2 = 2.0$ (no ECF from <b>3(b)(ii)</b> and <b>3(c)(i)</b> ) | <b>A1</b> |

9702/22

Cambridge International AS & A Level – Mark Scheme

October/November 2025

**PUBLISHED**

| Question | Answer | Marks |
|----------|--------|-------|
|          |        |       |